

Multi-omics and pharmacological characterization of patient-derived glioma cell lines

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Glioblastoma (GBM) is the most common brain tumor and remains incurable. Primary GBM cultures are widely used tools for drug screening, but there is a lack of genomic and pharmacological characterization for these primary GBM cultures. Here, we collect 50 patient-derived glioma cell (PDGC) lines and characterize them by whole genome sequencing, RNA sequencing, and drug response screening. We identify three molecular subtypes among PDGCs: mesenchymal (MES), proneural (PN), and oxidative phosphorylation (OXPHOS). Drug response profiling reveals that PN subtype PDGCs are sensitive to tyrosine kinase inhibitors, whereas OXPHOS subtype PDGCs are sensitive to histone deacetylase inhibitors, oxidative phosphorylation inhibitors, and HMG-CoA reductase inhibitors. PN and OXPHOS subtype PDGCs stably form tumors in vivo upon intracranial transplantation into immunodeficient mice, whereas most MES subtype PDGCs fail to form tumors in vivo. In addition, PDGCs cultured by serum-free medium, especially long-passage PDGCs, carry *MYC/MYCN* amplification, which is rare in GBM patients. Our study provides a valuable resource for understanding primary glioma cell cultures and clinical translation and highlights the problems of serum-free PDGC culture systems that cannot be ignored.

Glioblastoma (GBM) is the most common malignant tumor of the central nervous system^{1,2}. The clinical translation and treatment options for GBM remain a huge challenge³. Currently, the standard of care therapy for GBM is maximal surgical resection followed by temozolomide chemotherapy and adjuvant radiotherapy^{1,2,4}. The median survival time of GBM is only about 15 months, and this has not improved significantly in the past few decades^{1,5}.

Human cancer cell lines are the most commonly used models for mechanistic studies and drug screening in oncology research. Patient-derived glioma cells (PDGCs) cultured in serum-free neural stem cell

medium have been widely used in glioma studies^{6,7}. Compared to conventional glioma cell lines cultured in a serum-containing medium, PDGCs more faithfully recapture the mutational spectra and gene expression patterns of parental tumors^{8,9}. Several large-scale studies, including the Cancer Cell Line Encyclopedia (CCLE)^{10–12}, Genomics of Drug Sensitivity in Cancer (GDSC)¹³, and Cancer Therapeutics Response Portal (CTRP)¹⁴, have thoroughly characterized the genetic features and pharmacological responses of conventional glioma cell lines, while data depicting the genomic and pharmacological landscape for PDGCs cultured in serum-free medium are relatively scarce.

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In this work, we integrate whole genome sequencing (WGS), RNA sequencing (RNA-seq), and drug response data to characterize 50 PDGCs derived from GBM patients. Our data support the classification of the examined PDGCs into three molecular subtypes: mesenchymal (MES), proneural (PN), and oxidative phosphorylation (OXPHOS). Drug response profiles for 214 FDA-approved small molecules reveal that the PN subtype PDGCs are sensitive to tyrosine kinase inhibitors (TKIs), while the OXPHOS subtype PDGCs are sensitive to histone deacetylase (HDAC) inhibitors, oxidative phosphorylation inhibitors, and HMG-CoA reductase inhibitors. Notably, PDGCs cultured in the serum-free medium do preserve the genomic alterations, subtype identity, and informative intra-tumor heterogeneity of parental tissues, however, the serum-free culture system is associated with the *MYC/MYC*N amplification of cultured PDGCs. In addition, MES subtype GBM patients have a worse prognosis compared to the PN and OXPHOS subtypes, but we note that the majority of cultured PDGCs with MES subtype fail to establish tumors upon intracranial transplantation into immunodeficient mice. Our study provides an integrated resource to support glioma-related studies.

Results

Establishment and multi-omics profiling of 50 patient-derived glioma cell lines cultured in serum-free medium

To identify patient-derived glioma cells in detail, we collected and established 50 PDGCs derived from GBM (see “Methods” section). We performed RNA-seq on 50 PDGCs, whole exome sequencing (WES) on 10 PDGCs, and WGS on 49 PDGCs, as well as RNA-seq on 12 matched GBM tumor tissues of PDGCs (Fig. 1a and Supplementary Data 1).

To explore whether PDGCs recapitulate the genomic driver alterations of GBM^{9,15}, we analyzed the WGS data of PDGCs and revealed that PDGCs harbored genomic alterations including the amplification of the *PDGFRA* and *EGFR* genes, the deletion of the *PTEN* and *CDKN2A* genes, as well as mutations in the *TP53* and *NF1* genes (Fig. 1b). The overall alteration frequency of recurrent mutated genes in GBM patients showed high consistency between PDGCs and TCGA GBM cohort (Spearman $R = 0.68$, $p = 3.36 \times 10^{-5}$) (Supplementary Fig. 1a). It has been reported that glioma cell lines under serum-containing medium have difficulty in maintaining the amplification of *EGFR*¹⁶. Our PDGCs cultured in a serum-free medium maintained the amplification of *EGFR* (Supplementary Fig. 1b).

To determine whether PDGCs recapitulate the expression pattern of parental tissues, we performed RNA-seq on 12 PDGCs and their parental tissues. Wang et al.¹⁷ have identified a set of GBM intrinsically expressed genes. We selected the top 1000 most correlated GBM-intrinsic genes between PDGCs and GBM tissues (Supplementary Fig. 1c) to examine the expression pattern and found that PDGCs showed a high transcriptomic correlation with their derived GBM tissues (Spearman $R > 0.8$, Supplementary Fig. 1d). To eliminate the influence of non-tumor cells in the bulk tissue, we used RNA-seq data of tissues with high tumor purity (>90%) to perform principal component analysis (PCA) and found that PDGCs and their paired tissues had a natural inclination to each other (Supplementary Fig. 1e). The distances of the paired PDGC-tissue samples were significantly shorter than those of unpaired samples ($p = 0.00014$) (Supplementary Fig. 1f). These results collectively demonstrated that PDGCs cultured in serum-free medium faithfully replicate the genomic and transcriptomic features of GBM tissues.

Definition of transcriptional subtypes in PDGCs

According to the distinct transcriptional patterns, GBMs are classified into three subtypes: classical (CLS^{Wang}), proneural (PN^{Wang}), and mesenchymal (MES^{Wang}) at the bulk tumor level¹⁷. At the single-cell level, GBM tumor cells have been grouped into four cellular states: Oligodendrocyte progenitor-like (OPC-like), Neural progenitor-like (NPC-like), Astrocyte-like (AC-like), and Mesenchymal-like (MES-like)¹⁸.

These classification strategies based on GBM tissues are commonly used to categorize PDGCs^{6,19–21}, but no attempt has been made to classify PDGCs directly based on the PDGCs’ transcriptome. Seeking to categorize PDGCs’ subtypes in this study, we applied non-negative matrix factorization (NMF) to the RNA-seq data of the 50 PDGCs and found that the four-cluster option achieved the highest cophenetic score (Supplementary Fig. 2a, b). We identified differentially expressed genes and constructed a 100-gene signature for each cluster defined by NMF (Supplementary Data 2 and Fig. 1c).

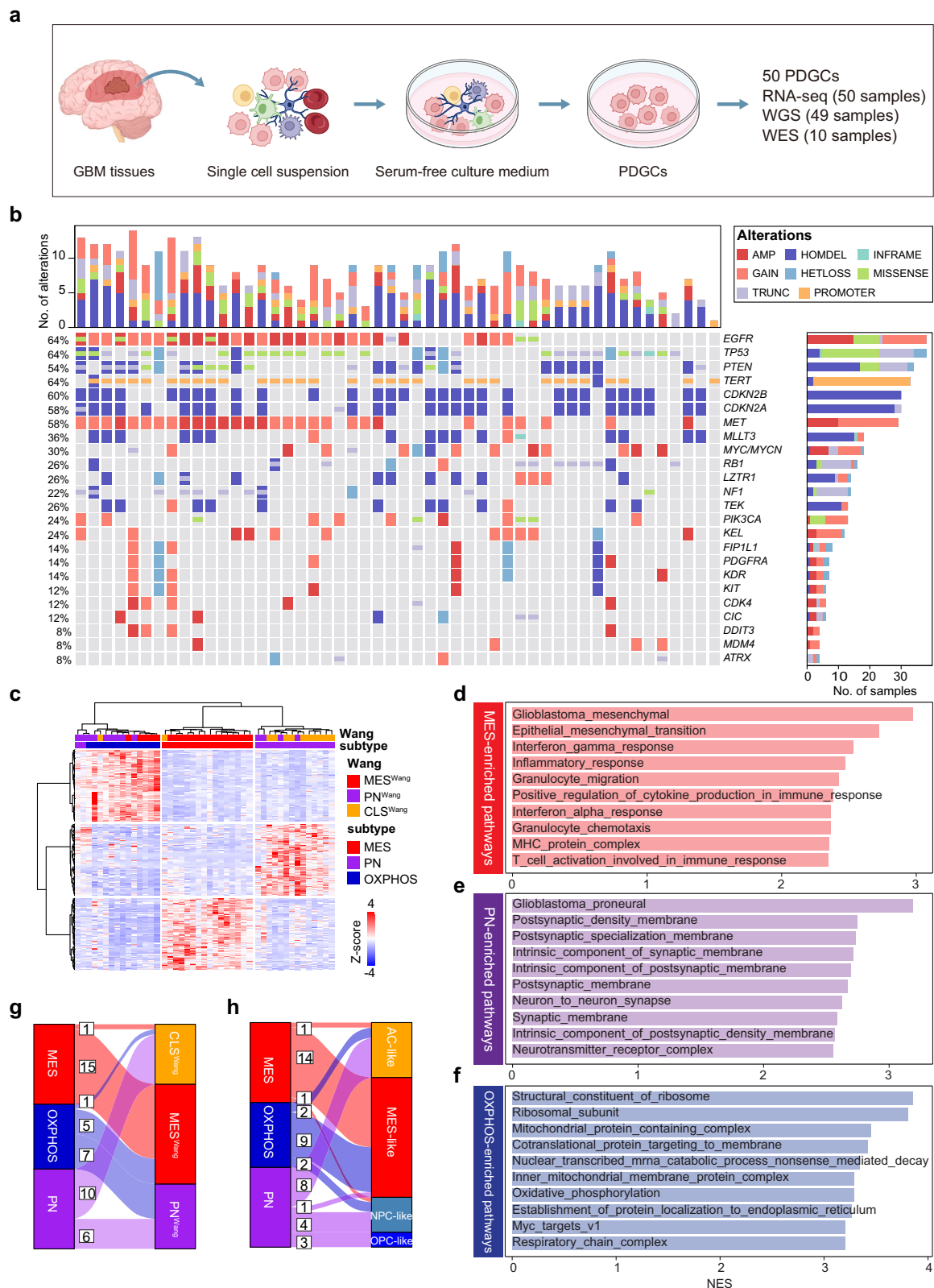
Gene set enrichment analysis (GSEA) revealed that PDGCs in cluster 1 ($n = 16$) were enriched for epithelial-mesenchymal transition (EMT) and several immune-associated pathways (Fig. 1d and Supplementary Fig. 2b), so we designated this cluster as the mesenchymal (MES) subtype. PDGCs in cluster 2 ($n = 16$) were enriched for neuron developmental-associated pathways (Fig. 1e and Supplementary Fig. 2b), and we designated this cluster as the proneural (PN) subtype. PDGCs in cluster 4 ($n = 13$) were enriched for mitochondria-associated functions and the oxidative phosphorylation pathway (Fig. 1f and Supplementary Fig. 2b), and we designated this cluster as the oxidative phosphorylation (OXPHOS) subtype. As there were only five cell lines in cluster 3, and we could not identify significantly differentially expressed gene signatures compared to the other clusters, we focused on the remaining three clusters. Garofano et al.²² identified a mitochondria (MTC) subtype. Both OXPHOS and MTC exhibited enrichment in the oxidative phosphorylation pathway. However, the OXPHOS subtype showed a clear association with pathways related to the cell cycle, including E2F targets and Myc targets (Supplementary Fig. 2c, d). To verify whether our subtypes were applicable to PDGCs from published studies, we examined the expression patterns of our signatures (Supplementary Data 2) in the PDGCs from Mack et al.²³ and HGCC datasets⁶ and found that our signatures were also differentially expressed in these PDGCs (Supplementary Fig. 2e).

To investigate the relationship between our subtypes and the two existing subtyping approaches^{17,18}, we mapped the subtype identities of PDGCs using all three subtyping approaches (Supplementary Data 2). Our MES subtype exhibited enrichment in the MES^{Wang} (15/16, 93.75%) and MES-like state (14/16, 87.5%), indicating that the mesenchymal signature was stable across the three subtyping approaches (Fig. 1g, h). Our PN subtype exhibited enrichment of PN^{Wang} (6/16, 37.5%) and CLS^{Wang} (10/16, 62.5%) and correlated with progenitor cell states (NPC- and OPC-like) (7/16, 43.75%). Our OXPHOS subtype mainly overlapped with PN^{Wang} (7/13, 53.85%) and MES^{Wang} (5/13, 38.46%) and correlated with MES-like cellular state (9/13, 69.23%) (Fig. 1g, h). These results were also observed in the other two independent datasets (Supplementary Fig. 2f, g).

To investigate the performance of our classification system in determining subtypes across PDGCs, we performed an in silico mimicry experiment based on the expression profiles of PDGCs. For each target PDGC, we randomly selected a background PDGC from the pool of remaining PDGCs and generated a mock expression profile of PDGC by blending the target PDGC with different proportions of the background PDGC (Supplementary Fig. 2h). Our classification system exhibited significantly less inconsistency in subtyping when compared to the classification systems proposed by Wang et al.¹⁷ and Garofano et al.²² (Supplementary Fig. 2i), indicating the robustness of our classification system in mitigating the influence of background noise.

PDGCs retain the subtype identity and heterogeneity of their parental tissues

To assess the extent to which PDGCs can retain the subtype identity of their parental tissues, we used RNA-seq data from 12 GBM tissues and their matched PDGCs to evaluate subtype changes upon culture. We found that 58.33% (7/12) of PDGCs retained the subtype of their parental tissues after culture. The five exceptions were one MES tissue transformed into OXPHOS-PDGC, one MES tissue transformed into PN-



PDGC, one OXPHOS tissue transformed into MES-PDGC, and two PN tissues transformed into MES-PDGCs (Fig. 2a and Supplementary Data 3). A similar trend was observed when we examined another published dataset²³, wherein 60% (6/10) of PDGCs retained the subtype of their parental tissues, with one OXPHOS tissue transformed into PN-PDGC, and three OXPHOS tissues transformed into MES-PDGCs (Fig. 2b and Supplementary Data 3). These two lines of evidence

both supported that most PDGCs retained the subtype identity of their parental tissues.

GBM is well known for its heterogeneity, and previous single-cell RNA-seq (scRNA-seq) analyses have demonstrated the co-occurrence of multiple tumor cell subtypes within a single tumor tissue^{18,24}. To evaluate the extent to which cultured PDGCs retained the heterogeneity of parental tumor tissues, we analyzed scRNA-seq data from

Fig. 1 | Definition of transcriptional subtypes for PDGCs. **a** Schematic diagram showing the generation of PDGCs. GBM tissues from patients were dissected and digested into single-cell suspension, which was then cultured in a serum-free medium to obtain PDGCs. A total of 50 PDGCs were collected or generated in this study, of which 50 PDGCs underwent RNA-seq, 10 PDGCs underwent WES, and 49 PDGCs underwent WGS. **b** Oncoprint showing the genomic alterations of 50 PDGCs. HOMDEL: homozygous deletion, AMP: high-level amplification, INFRAME: in-frame insertion and deletion, MISSENSE: missense mutation, PROMOTER: mutation localized in promoters, TRUNC: truncation mutation, GAIN: low-level

copy number gain, HETLOSS: heterozygous deletion. **c** Heatmap showing the expression levels of subtype-specific signatures. Gene expression values were normalized by z-score. The rows represent genes, and the columns represent PDGCs. **d–f** Barplots showing the enriched pathways by MES (**d**), PN (**e**), and OXPHOS (**f**) subtype. Bars were colored according to subtype identity. **g, h** Sankey plots showing subtype assignment ($n = 45$) change flow. The left columns represent subtypes defined by us and the right columns represent subtypes defined by Wang et al. ¹⁷ (**g**) or by Neftel et al. ¹⁸ (**h**). Source data are provided as a Source Data file.

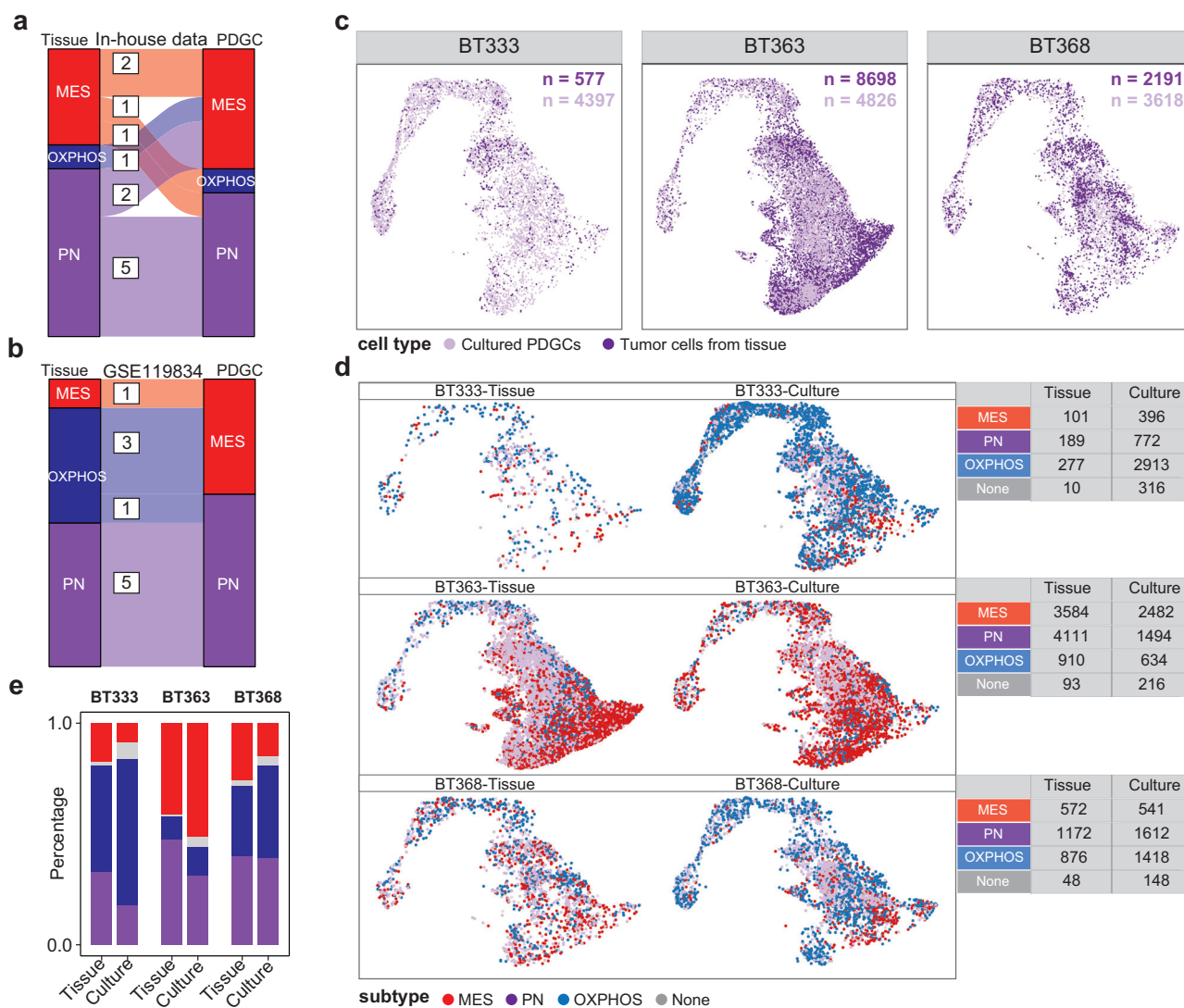


Fig. 2 | Subtype of cultured PDGCs and their original tissues. **a, b** Sankey plots showing the change in subtype assignment between cultured PDGCs and their parental tissues using in-house data ($n = 12$) (**a**) and previously published data GSE119834 ($n = 10$) (**b**). **c** Uniform Manifold Approximation and Projection (UMAP) visualization of GBM tumor cells from tissue and cultured PDGCs, colored by cell

type. **d** UMAP visualization of GBM tumor cells from tissues and cultured PDGCs, colored by subtype assignment. Tables next to the UMAP visualization summarizing the number of cells of each subtype. **e** The percentage of each subtype in GBM tumor cells from tissues and cultured PDGCs. Source data are provided as a Source Data file.

the study of Couturier et al., which assessed both cultured PDGCs and their corresponding tumor tissues²⁵. We initially filtered out non-malignant cells and then re-clustered the tumor cells for the heterogeneity analysis (Supplementary Fig. 3). Uniform Manifold Approximation and Projection (UMAP) embedding indicated that the GBM cells isolated from the tissues were clustered with the cultured PDGCs, supporting the transcriptional similarity between the two sample types (Fig. 2c). Similar to tumor tissues, cultured PDGCs derived from a single tumor tissue contained different subtypes of cells (Fig. 2d).

We subsequently compared the cell composition of different subtypes between the cultured PDGCs and their parental tissues. For patient BT333, the proportion of cells with MES and PN subtypes decreased, whereas the proportion of cells with the OXPHOS subtype increased in the cultured PDGCs (tumor tissue vs. cultures PDGCs: MES, 17.5% vs. 9.0%; PN, 32.8% vs. 17.6%; OXPHOS, 48.0% vs. 66.2%). Similar trends of decreasing tumor cells of the MES and PN subtypes and increasing tumor cells of the OXPHOS subtype were observed in the PDGCs of patient BT363 (tumor tissue vs. cultures PDGCs: MES,

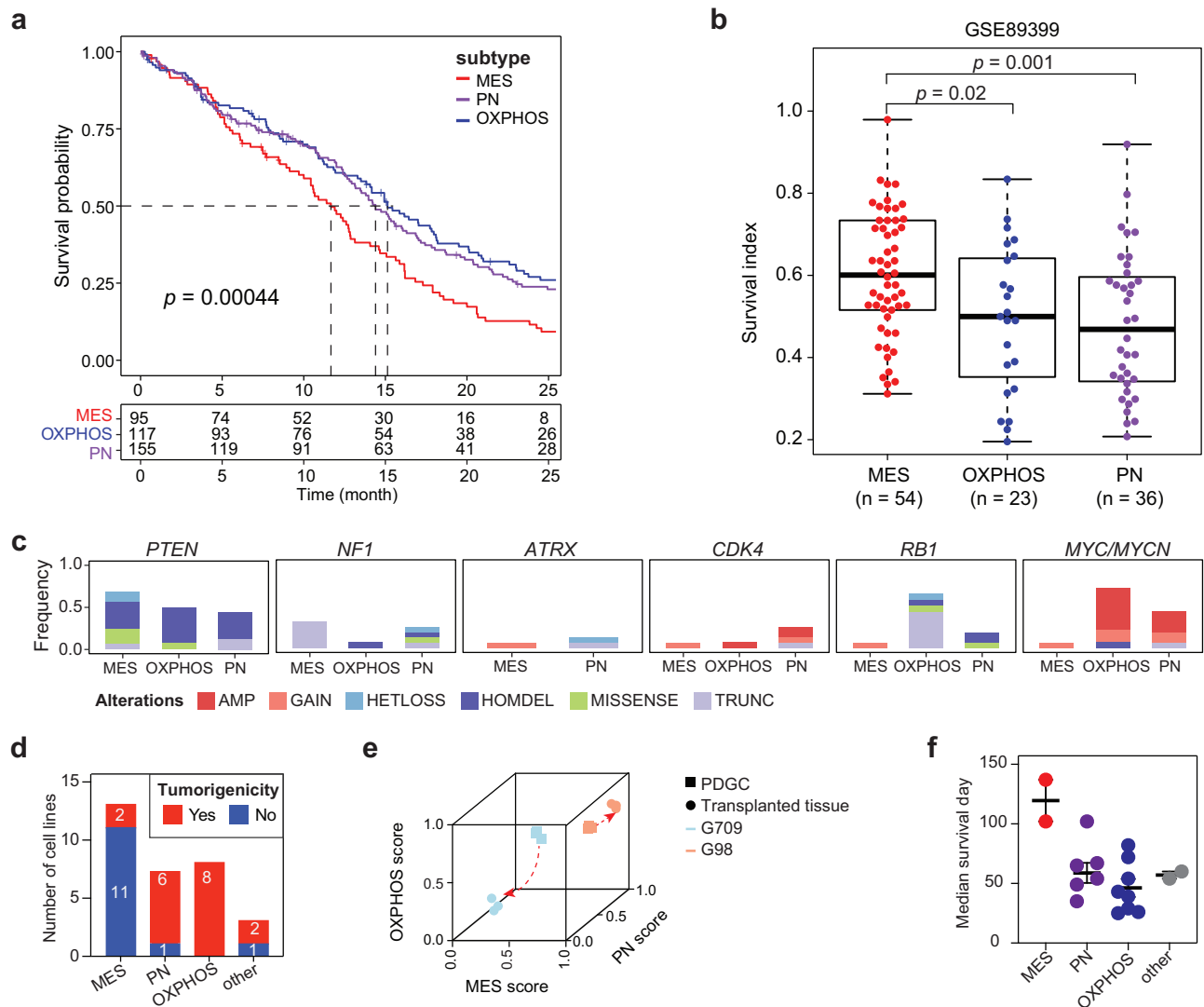


Fig. 3 | Clinical, genomic, and tumorigenic comparisons of defined subtypes. **a** Kaplan–Meier survival curves for GBM patients from the TCGA cohort with different subtypes. The P -value was calculated using the log-rank test. **b** Response to radiotherapy for defined subtypes. The y-axis represents the percentage of surviving cells upon radiation. P -values were calculated using the two-sided Wilcoxon rank-sum exact test. Boxplots show the median (center line), the upper and lower quantiles (box), and the range of the data (whiskers). **c** Frequency of genomic alterations of the indicated genes in PDGCs of different subtypes. Bars were colored by alteration types. AMP: high-level amplification, GAIN: low-level gain, HETLOSS:

heterozygous deletion, HOMDEL: homozygous deletion, MISSENSE: missense mutation, TRUNC: truncation mutation. **d** Stacked barplots showing the tumorigenicity of PDGCs of different subtypes. For each cell line, 5×10^5 cells were injected into the brains of nude mice. **e** 3D scatterplot showing the subtype score of PDGCs and their transplanted tumors' cell culture. **f** Scatter plots showing the median survival time of nude mice after injection of PDGCs of different subtypes. MES ($n = 2$), PN ($n = 6$), OXPPOS ($n = 8$), other ($n = 2$). The lines were plotted as mean $\pm$ SEM. Source data are provided as a Source Data file.

41.2% vs. 51.4%; PN, 47.3% vs. 31.0%; OXPPOS, 10.5% vs. 13.1%) and BT368 (tumor tissue vs. cultures PDGCs: MES, 21.4% vs. 14.5%; PN, 43.9% vs. 43.3%; OXPPOS, 32.8% vs. 38.1%) (Fig. 2e). These results demonstrate that although the cultured PDGCs retain the multi-subtype tumor heterogeneity of the GBM parental tissues, their subtype composition changes slightly.

Clinical, genomic, and tumorigenic comparisons of PDGCs with different subtypes

To investigate the relationship between our classification system and the subtyping approach defined by Wang et al.¹⁷ for GBM tissues, we classified GBM samples from the TCGA project using both subtyping strategies. Similar to what we had observed in PDGCs, our MES subtype exhibited enrichment in MES^{Wang} subtype (69/88, 78.41%), our PN subtype exhibited enrichment in PN^{Wang} (62/146, 42.47%), and CLS^{Wang} (80/146, 54.79%), while our OXPPOS subtype was distributed

in all three subtypes defined by Wang et al.¹⁷ (Supplementary Fig. 4a, b). Additionally, survival analysis showed that GBM patients of the MES subtype had significantly shorter overall survival (OS) compared to patients of the other two subtypes (median OS: 11.7 months for the MES subtype, 14.4 months for the PN subtype, and 15.1 months for the OXPPOS subtype, log-rank test, $p = 0.00044$) (Fig. 3a).

To assess the apparent sensitivity of the three subtypes we defined to radiotherapy, we used the dataset from the study of Segerman et al.²⁶, wherein they established a library of primary glioma cell clones and examined their gene expression as well as sensitivity to radiation therapy. After assigning a subtype identity to the glioma cell clones, we found that the MES subtype cell clones appeared to be more resistant to radiotherapy (median survival index: 0.60), while the PN subtype cell clones were relatively sensitive (median survival index: 0.47) (Fig. 3b).

To identify the genomic alterations in our defined subtypes, we detected copy number variations (CNVs) and mutations by analyzing WGS data of PDGCs (Fig. 3c and Supplementary 4c). This analysis revealed that *PTEN* (MES: 68.8%, PN: 43.8%, OXPPOS: 50%) and *NFI* (MES: 31.2%, PN: 25%, OXPPOS: 7.14%) genomic alterations were enriched in the MES subtype, *ATRX* (MES: 6.25%, PN: 12.5%) and *CDK4* (MES: 6.25%, PN: 25%, OXPPOS: 7.14%) genomic alterations were enriched in the PN subtype, while *RBI* (MES: 6.25%, PN: 18.8%, OXPPOS: 64.3%) and *MYC/MYCN* (MES: 6.25%, PN: 43.8%, OXPPOS: 71.4%) genomic alterations were enriched in the PN and OXPPOS subtypes (Fig. 3c). When reviewing the CNVs and somatic mutations of GBM patients from TCGA, we found a similar trend of genomic alteration distribution among our three subtypes (Supplementary Fig. 4c).

To assess the tumorigenic capacity of different subtypes of PDGCs, we transplanted 5×10^5 PDGCs into the brains of immunodeficient mice. Most PDGCs of the PN (6/7) and OXPPOS (8/8) subtypes were able to form intracranial tumors in nude mice (Fig. 3d). Although GBM patients of the MES subtype had the shortest overall survival compared to those of the PN and OXPPOS subtypes (Fig. 3a), PDGCs of the MES subtype had the lowest tumorigenicity (MES: 15.38%, PN: 85.71%, OXPPOS: 100%). Specifically, of the 13 MES-PDGCs tested, only a fraction of mice transplanted with two cell lines (G709 and G98) formed intracranial tumors within 6 months after inoculation (Fig. 3d and Supplementary Data 4). We performed RNA-seq on cell cultures derived from the transplanted tumor of G98 and G709 and found that the subtype of G709 changed from MES to PN (Fig. 3e). Furthermore, tumor-bearing mice transplanted with MES subtype PDGCs tended to have longer median survival time than those transplanted with non-MES subtype PDGCs (MES: 119.5 days, non-MES: 54 days) (Fig. 3f). Collectively, these results showed differences among the three subtypes in terms of apparent radiotherapy sensitivity, overall survival, genomic alterations, and tumorigenic capacity.

PDGCs with different subtypes have distinct drug responses

To explore the drug response of PDGCs with different subtypes, we initially evaluated the effects of 1,466 FDA-approved small molecules on the cell viability of three PDGCs (G98, G118, and G709) (Supplementary Table 1). This screening yielded 214 small molecules that had at least a 75% inhibitory effect on at least one PDGC at a concentration of 10 μ M (Fig. 4a). We subsequently measured the cell viability of 41 PDGCs treated with these 214 compounds (at a dose of 5 μ M) (Supplementary Fig. 5 and Supplementary Data 5). We defined normalized cell viability to measure the inhibitory effect of a drug on PDGCs. The lower the normalized cell viability, the stronger the inhibitory effect of the drug on the cells (Supplementary Data 6).

Cell viability analysis showed that the response of PDGCs to drugs was variable, with the median of normalized cell viability ranging from 0.037 to -9.76 (Supplementary Fig. 5). We found that 96 of the drugs exerted little inhibitory effect on any PDGCs (normalized cell viability: mean > -1 or standard deviation < 0.5), so we retained the remaining 118 drugs for further analysis (Fig. 4b). Overall, PDGCs with MES subtype were more resistant to drugs (mean normalized cell viability: -2.24), compared to PDGCs with PN and OXPPOS subtypes (mean normalized cell viability for PN: -3.24, OXPPOS: -3.09) (Fig. 4c).

To capture the drug response patterns across PDGC subtypes, we performed an unsupervised cluster analysis of cell viability profiles (Fig. 4b). Drugs in cluster 1 were able to inhibit cell growth across three PDGC subtypes, especially the PN subtype (Fig. 4b), such as antibiotic drug Cloxiquine²⁷ (mean normalized cell viability: MES, -3.09; PN, -8.59; OXPPOS, -4.60), the dopamine receptor blocker penfluridol²⁸ (mean normalized cell viability: MES, -5.04; PN, -8.74; OXPPOS, -5.78), and the TKI ponatinib²⁹ (mean normalized cell viability: MES, -3.67; PN, -7.36; OXPPOS, -4.54) (Fig. 4d). In addition, PDGCs with PN subtype were sensitive to drugs such as TKIs (foretinib³⁰ and entrectinib³¹), antibiotic drugs (cetrimeronium bromide³² and

chlorhexidine 2HCl^{33,34}), and antihistamine terfenadine³⁵ (Fig. 4d and Supplementary Data 7). PDGCs with OXPPOS subtype were sensitive to drugs including inhibitors of mitochondrial activity (Terfenadine³⁶ and Ebastine³⁷), HDAC inhibitors³⁸ (panobinostat, belinostat, pracinostat and abexinostat), and HMG-CoA reductase inhibitors³⁹ (pitavastatin, simvastatin, and lovastatin) (Fig. 4e and Supplementary Data 7). These results reveal that PDGCs of different subtypes showed differences in drug response.

PDGCs of the OXPPOS subtype are sensitive to HMG-CoA reductase inhibitor lovastatin

Our drug screening showed that OXPPOS subtype PDGCs tended to be more sensitive to HMG-CoA reductase inhibitors than PN or MES subtype PDGCs (Fig. 4e). To further confirm this observation, we treated the MES subtype (GBM13 and G755_5), PN subtype (MGG8 and GSC23), and OXPPOS subtype (BNI423 and BNI274) PDGCs with 1 μ M lovastatin. The expression of cholesterol synthesis genes (*HMGCS1*, *HMGCR*, and *SQLE*)^{40,41} in all PDGCs increased significantly compared to the DMSO group (Supplementary Fig. 6a). The OXPPOS subtype PDGCs had the lowest IC₅₀ value among the subtypes (mean IC₅₀: MES, 36.92 μ M; PN, 39.99 μ M; OXPPOS, 0.39 μ M) (Supplementary Fig. 6b and Supplementary Data 8). An annexin V/PI assay of the OXPPOS subtype PDGC BIN423 showed that treatment with 1 μ M lovastatin resulted in a significant increase in the number of apoptotic cells (Supplementary Fig. 6c), whereas this was not the case for the PN subtype PDGC CB1838 and the MES subtype PDGCs (GBM13 and CB1018) (Supplementary Fig. 6d). These results indicated that although cholesterol synthesis pathways were inhibited in all PDGCs, the OXPPOS subtype was more sensitive to lovastatin treatment than the PN and MES subtypes.

Statins block the synthesis of cholesterol by inhibiting HMG-CoA reductase (Supplementary Fig. 6e). We found that supplementation of cholesterol (or its precursor lanosterol) in lovastatin-treated BIN423 cells abolished the inhibitory effect of lovastatin (Fig. 5a, b, and Supplementary Fig. 6f), suggesting that lovastatin inhibits the growth of OXPPOS subtype PDGCs by reducing cholesterol level.

To investigate the downstream mechanism of lovastatin treatment to OXPPOS subtype PDGCs, we performed RNA-seq analysis on OXPPOS subtype PDGCs (BNI274, BNI423, and BNI7_11) with or without 1 μ M lovastatin treatment. Gene set enrichment analysis showed that oxidative phosphorylation pathway activity was significantly inhibited upon 18 h of lovastatin treatment (NES = -2.199) (Fig. 5c). This observation prompted us to test whether lovastatin treatment to OXPPOS subtype PDGCs would influence the number of mitochondria. We visualized the mitochondria using MitoTracker and found that the number of mitochondria was significantly reduced in the lovastatin-treated BNI423 cells compared with the vehicle control group and this defect can be rescued by cholesterol supplementation (Fig. 5d).

In agreement with the RNA-seq analysis, we also found that treatment of BNI423 with lovastatin (Student's t-test, $p = 0.003$) or with the mitochondrial complex I inhibitor rotenone⁴² (Student's t-test, $p = 0.0002$) resulted in a significant increase in ADP/ATP ratio compared with their respective vehicle controls (Fig. 5e). Again, the phenotype of reduced ATP production upon lovastatin treatment was rescued by cholesterol supplementation (Student's t-test, $p = 0.0007$) (Fig. 5f). Thus, lovastatin treatment to an OXPPOS subtype PDGC culture lowered cholesterol levels, thereby reducing the number of mitochondria and the activity of oxidative phosphorylation pathway. Daily oral gavage of lovastatin significantly reduced the tumor size of subcutaneously transplanted tumors (OXPPOS subtype PDGC BNI17) (Fig. 5g and Supplementary Fig. 6g), but this regime was unable to prohibit the growth of intracranial tumors (Supplementary Fig. 6h). Intracranial infusion of lovastatin via osmotic pumps (42 μ g/day/mouse) significantly reduced mitotic index (Fig. 5h and Supplementary Fig. 6i) (Student's t-test, $p = 0.038$) and increased the intensity of

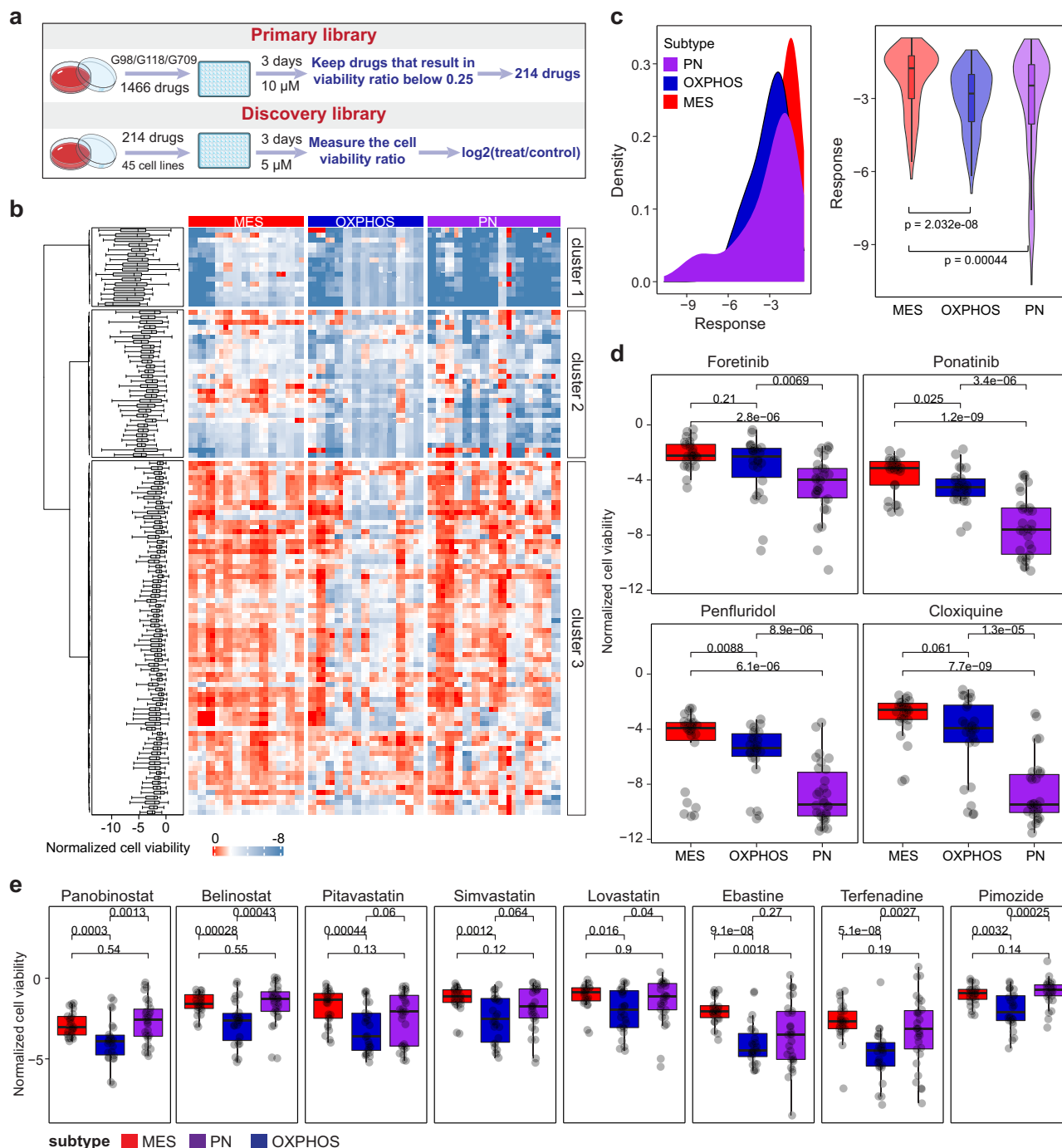


Fig. 4 | Drug responses of PDGCs with different subtypes. **a** Schematic diagram showing the drug screening process. **b** Heatmap showing the drug response of PDGCs with different subtypes. The rows represent drugs, and the columns represent cell lines, colored according to the cell viability upon the treatment of corresponding drugs. The boxplots on the left side show the response of cell lines to specific drug treatments. **c** Drug ($n = 118$) response of PDGCs with different subtypes. P -values were calculated by a two-sided Wilcoxon rank-sum exact test.

d, e Normalized cell viability of PDGCs with different subtypes upon treatment with the indicated drugs. Representative drugs that effectively inhibited the growth of PN and OXPPOS subtype PDGCs are listed in **(d)** and **(e)**, respectively. MES ($n = 26$), PN ($n = 30$), OXPPOS ($n = 26$). P -values were calculated by a two-sided Wilcoxon rank-sum exact test. In **b–e**, boxplots show the median (center line), the upper and lower quantiles (box), and the range of the data (whiskers). Source data are provided as a Source Data file.

cleaved Caspase-3 staining (Fig. 5i and Supplementary Fig. 6i) (Student's t -test, $p = 0.000003$) of transplanted brain tumors (OXPPOS subtype PDGC BNI423).

Serum-free culture system is associated with *MYC/MYC*N amplification

Of the 50 PDGCs in this study, 17 (34%) showed chromosomal and/or extrachromosomal DNA (ecDNA) amplification of *MYC/MYC*N

(Supplementary Data 1). To investigate whether *MYC/MYC*N amplification also occurs in PDGCs cultured in serum-free media used in other studies, we analyzed the DNA methylation data of PDGCs⁴³ to infer copy number and found that the copy numbers of *MYC/MYC*N in these cells showed an amplification trend, even though most of them had been passaged less than three generations (Supplementary Fig. 7a). To explore whether the activity of the MYC pathway is regulated by the cell culture system, we compared the transcriptomes of PDGCs and the

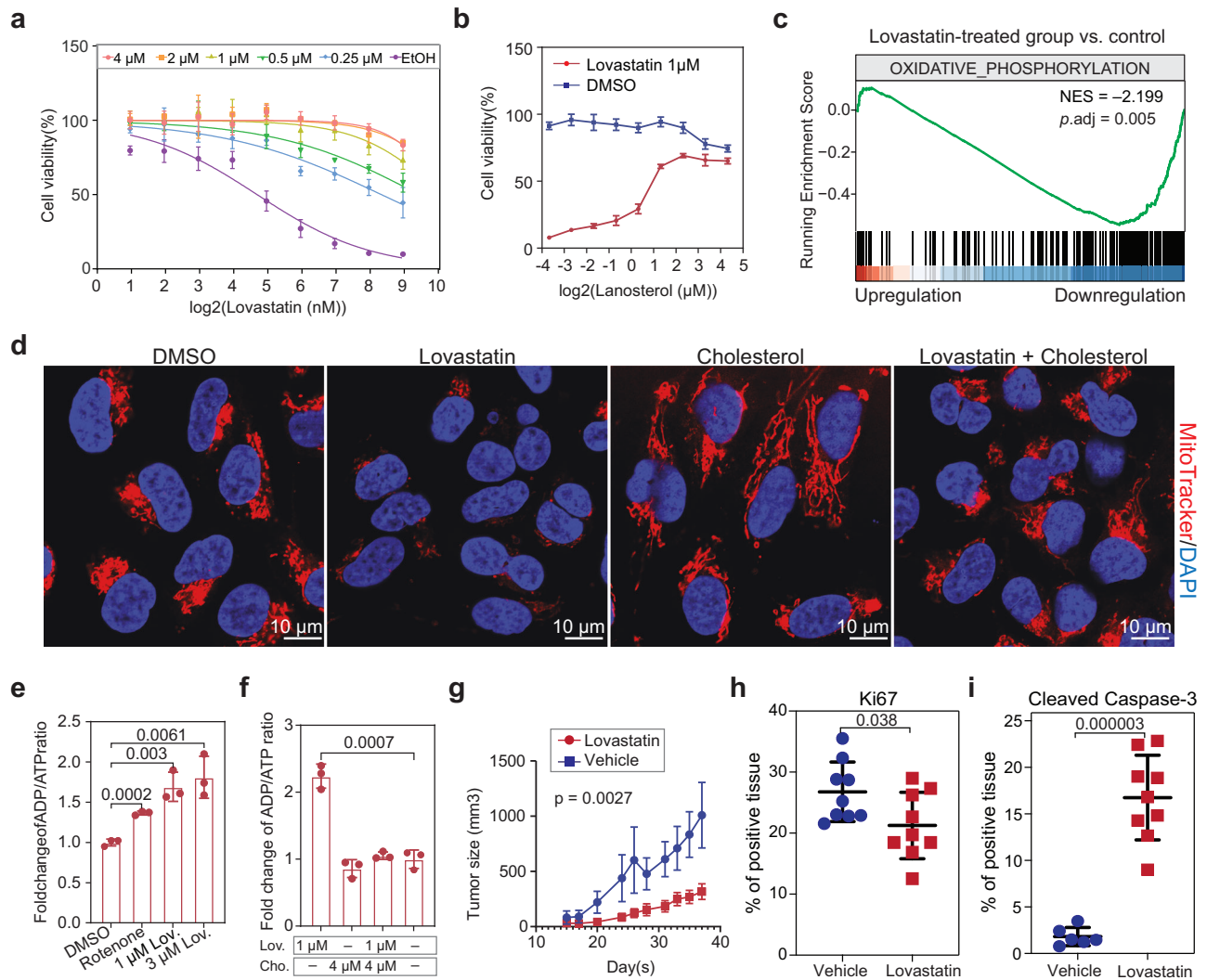


Fig. 5 | Lovastatin inhibits the growth of OXPHOS subtype PDGCs by attenuating mitochondrial respiration. **a** Cell viability of lovastatin-treated BN1423 supplemented with different concentrations of cholesterol (biological replicates $n = 3$). EtOH: ethyl alcohol. Data are presented as mean $\pm$ SD. **b** Cell viability of lovastatin-treated BN1423 supplemented with different concentrations of lanosterol (biological replicates $n = 3$). Data are presented as mean $\pm$ SD. **c** GSEA plot of OXPHOS subtype PDGCs treated with lovastatin compared to PDGCs without lovastatin treatment. GSEA is performed by R package clusterProfiler. P -value was calculated by the two-sided Kolmogorov-Smirnov test and adjusted by FDR. **d** Representative MitoTracker staining of BN1423 under different treatments for 16 h. The mitochondria were stained with MitoTracker (red), and the nucleus was stained with DAPI (blue). Scale bar, 10 μ m. **e** Relative cellular ADP/ATP ratio of BN1423 under different treatments (biological replicates $n = 3$). The mitochondrial complex I inhibitor rotenone was used as a positive control. P -values were

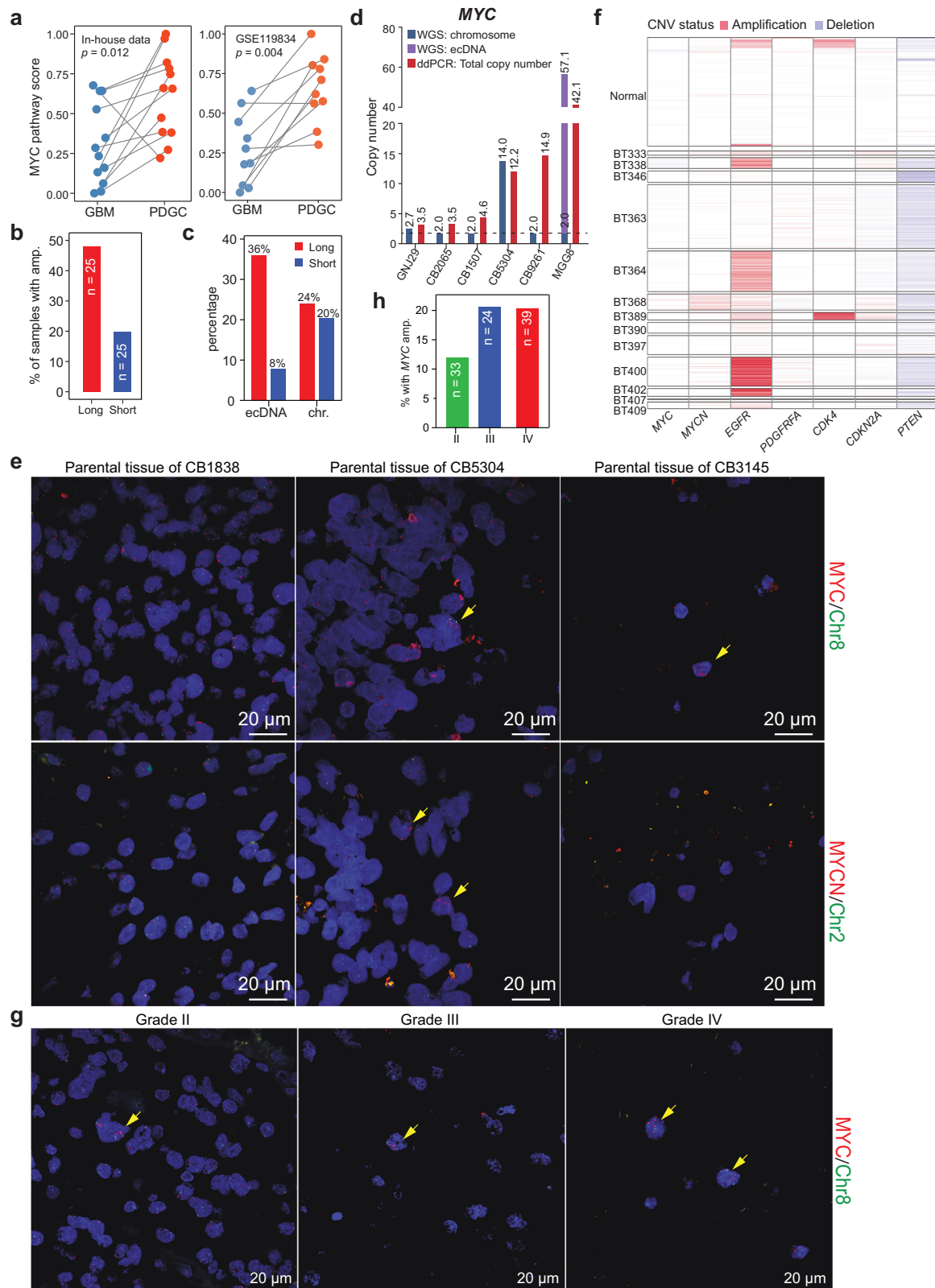
calculated by the two-sided Student's t -test. Lov.: Lovastatin. Data are presented as mean $\pm$ SD. **f** Relative cellular ADP/ATP ratio of lovastatin-treated BN1423 with or without cholesterol supplementation (biological replicates $n = 3$). P -values were calculated using the two-sided Student's t -test. Lov.: Lovastatin, Cho.: Cholesterol. Data are presented as mean $\pm$ SD. **g** Tumor size of mice with subcutaneously inoculated BN117 ($n = 6$) treated with lovastatin or vehicle. P -values were calculated using the two-sided Student's t -test. Data are presented as mean $\pm$ SEM. **h**, **i** Quantification of Ki67+ (**h**) and cleaved Caspase-3-stained (**i**) brain tumor tissues. BN1423 were transplanted into the brains of nude mice. After five weeks, lovastatin was administered by brain infusion (42 μ g per day per mouse) for three days, and the whole brains were harvested for histopathologic analysis. P -values were calculated using the two-sided Student's t -test. Data are presented as mean $\pm$ SD. In (**h**), Vehicle ($n = 9$), Lovastatin ($n = 9$). In (**i**), Vehicle ($n = 6$), Lovastatin ($n = 9$). Source data are provided as a Source Data file.

corresponding tumor tissues and found that the level of the MYC pathway increased significantly after culture (Fig. 6a). We also observed a similar phenomenon in another independent dataset²³ (Fig. 6a).

MYC/MYC amplification is not common in GBM patients. According to data from the TCGA project, only about 3.8% of GBM patients have MYC/MYC amplification¹⁵. To explore whether PDGCs cultured in serum-containing media exhibit MYC/MYC amplification, we analyzed the WGS data of 280 tumor cell lines cultured under serum-containing conditions in the CCLC project¹⁰. MYC amplification is frequently observed in lung adenocarcinoma (LUAD), lung squamous cell carcinoma (LUSC), breast invasive carcinoma (BRCA),

stomach adenocarcinoma (STAD), and ovarian cancer (OV)⁴⁴. In consistent with this view, frequent MYC amplification was observed in the corresponding cell lines (percentage of MYC amplification in cell lines: Lung cancer, 20.41%; Breast cancer, 22.86%; Gastric cancer, 19.05%; Ovarian cancer, 17.39%). No MYC amplification was detected in 15 central nervous system tumor cell lines (Supplementary Fig. 7b). These results suggest that the MYC/MYC amplification observed in glioma cell lines may be related to the culture conditions.

Of the 50 PDGCs in this study, 24 PDGCs were subjected to WGS at a short-passage (<10 passages) stage, while others had been passed through multiple passages (>10 passages) (Supplementary Data 1). To further verify the relationship between MYC/MYC amplification and



PDGC culture system, we compared the frequency of *MYC/MYC* amplification in short-passage and long-passage (>10 passages) PDGCs using their WGS data. We found that the frequency of *MYC/MYC* amplification events was significantly higher in long-passage PDGCs than in short-passage PDGCs (48% vs. 20%, Fisher's exact test, $p = 0.036$) (Fig. 6b). Meanwhile, we observed that 36% of long-passage PDGCs exhibited *MYC/MYC* ecDNA amplification, whereas this

frequency was only 8% in short-passage PDGCs (Fig. 6c). When PDGCs passed more than 20 generations, we quantified their copy number of *MYC* by droplet digital PCR (ddPCR) and compared them with the copy number of *MYC* determined by WGS. PDGCs CB1507 and CB9261 did not carry *MYC* amplification at the short-passage stage according to their WGS data and became *MYC*-amplified cell lines after prolonged passage according to the ddPCR results (Fig. 6d and Supplementary

Fig. 6 | MYC/MYCIN amplification in cultured PDGCs. **a** MYC pathway expression score in PDGCs and matched GBM tissues. In-house data ($n = 12$), GSE119834 ($n = 10$). The P -value was calculated by a two-sided paired Wilcoxon rank-sum exact test. **b** Percentage of samples with MYC/MYCIN amplification in long- ($n = 25$) and short-passage ($n = 25$) PDGCs. Cell lines cultured more than 10 passages were categorized as long-passage, otherwise, the cell lines were categorized as short-passage. The P -value was calculated by one-sided Fisher's exact test, $p = 0.036$. **c** Percentage of samples with MYC/MYCIN DNA amplification or with MYC/MYCIN ecDNA amplification. The red bar indicates long-passage PDGCs, while the blue bar indicates short-passage PDGCs. Chr.: chromosomal DNA, ecDNA: extra-chromosomal DNA. **d** The copy number of MYC. The blue bars represent the chromosomal copy number detected by WGS, the purple bars represent the ecDNA copy number detected by WGS, and the red bars represent the total copy number detected by ddPCR. **e** Representative FISH images showing the MYC/MYCIN

amplification in primary GBM tissues from patients CB1838, CB5304, and CB3145. The yellow arrow indicates cells with MYC/MYCIN amplification. Scale bar, 20 μm . A total of 13 tissues were subject to FISH, of which 9 samples without MYC/MYCIN amplification, 3 samples with MYC amplification, and 1 sample with MYCIN amplification. **f** Heatmap showing the CNV of the indicated genes, colored by CNV status (red for amplification and blue for deletion). The rows represent cells isolated from GBM patients and the columns represent the genomic locations. CNV levels were inferred by R package infercnv. **g** Representative FISH images showing rare MYC-amplifying tumor cells in grades II, III, and IV glioma tissues. The yellow arrow indicates cells with MYC amplification. Scale bar, 20 μm . A total of 96 glioma tissues were tested, with grade II ($n = 33$), III ($n = 24$), and IV ($n = 39$). **h** Percentage of samples harboring more than three MYC amplification cells in grade II ($n = 33$), III ($n = 24$), and IV ($n = 39$) glioma tissues. Source data are provided as a Source Data file.

7c), confirming the MYC amplification was associated with the serum-free PDGC culture system.

Since MYC/MYCIN amplification was also observed in low-passage PDGCs, we speculated that a small number of MYC/MYCIN-amplified cells might be present in the original GBM tissue. To verify this, we performed fluorescence in situ hybridization (FISH) detection on the matched tumor tissues of the PDGCs. Our WGS analysis of PDGCs showed that CB5304 and CB3145 had MYC or MYCIN amplification, while CB1838 had no MYC or MYCIN amplification (Supplementary Data 1). FISH examination revealed that MYC/MYCIN-amplified cells were present in the initial tissues of CB5304 and CB3145, but not in the initial tissue of CB1838 (Fig. 6e).

To verify the presence of rare MYC/MYCIN-amplified tumor cells, which cannot be detected at the bulk level in glioma samples, we first analyzed the scRNA-seq data of GBM tissue from Couturier et al.²⁵ and obtained 31,159 tumor cells after quality control. The CNVs inferred from the scRNA-seq data showed that 38.8% of tumor cells had epidermal growth factor receptor (*EGFR*) amplification and 50.8% had *PTEN* loss. Remarkably, we found that 1.2% and 5.4% of tumor cells exhibited MYC and MYCIN amplification, respectively (Fig. 6f). When we analyzed the scRNA-seq data of GBM tissue from Neftel et al.¹⁸, we detected MYC amplification in 4.6% of tumor cells (Supplementary Fig. 7d). We also examined an additional 96 glioblastoma tissue samples (33 grade II, 24 grade III, and 39 grade IV) using FISH. We detected cells with MYC amplification, and these cells tended to cluster together (Fig. 6g). In addition, high-grade glioblastoma tissues were more likely to carry MYC-amplified cells than low-grade glioblastoma tissues (grade II, 12.12%; grade III, 20.83%; grade IV, 20.51%) (Fig. 6h). Overall, these findings support the existence of rare MYC-amplified cells in glioma tissues.

Discussion

In this study, we integrated genomic, transcriptomic, and drug response profiles to characterize the patient-derived GBM cell lines. Our analysis classified the 50 PDGCs into three subtypes: MES, PN, and OXPPOS. Notably, the PN subtype PDGCs demonstrated sensitivity to TKIs, while the OXPPOS subtype PDGCs were sensitive to HDAC inhibitors, oxidative phosphorylation inhibitors, and HMG-CoA reductase inhibitors.

Prior to our study, several studies have established and characterized primary glioma cell lines^{6,19,20,23}. These studies used the strategies developed by Verhaak et al.⁴⁵ or Wang et al.¹⁷ to determine the subtype identity of the cell lines. Both strategies generated subtype-specific features based on the transcriptome of patient tissues, which contained non-malignant cells and did not accurately reflect the expression profile of pure tumor cells. Based on the transcriptome of PDGCs, we determined the MES, PN, and OXPPOS subtypes, an identification strategy that excluded contamination by non-malignant cells and had the same predictive power for clinical prognosis as previous tissue-based methods. The MES subtype was strongly

correlated with the MES^{Wang} subtype, suggesting that mesenchymal features are stable in GBM cells. The PN subtype was enriched in both the PN^{Wang} and CLS^{Wang} subtypes. Our subtypes were validated in GBM cell lines from the Mack²³ and Johansson datasets⁶ (Supplementary Fig. 2e). Studies using GBM mouse models have demonstrated the plasticity of subtypes between CLS^{Wang} and PN^{Wang}⁴⁶, which were unified into PN subtype in our subtype system. The OXPPOS subtype is characterized by high expression of genes involved in mitochondrial oxidative phosphorylation. The MTC subtype defined by Garofano et al.²² also enriched the oxidative phosphorylation pathway, but only two genes overlapped between the signatures of the MTC and OXPPOS subtypes, indicating their distinctive transcriptomic features.

Our drug response analysis showed that multiple drugs have subtype-selective inhibitory effects on glioma cell growth. For example, TKIs were particularly effective against PN subtype PDGCs, consistent with a recent study showing that the multi-target TKI dasatinib has therapeutic effects on oligodendroglioma-derived GBMs⁴⁷. Oxidative phosphorylation inhibitors were particularly effective against OXPPOS subtype PDGCs. In contrast to PN and OXPPOS subtype PDGCs, MES subtype PDGCs were less sensitive to the investigated drugs, which is consistent with the poorer clinical prognosis of MES subtype GBM patients. As cell culture models are widely used for screening various anti-tumor drugs, transcriptome-based subtyping may help to classify drug response more precisely. For example, although lovastatin inhibits the cholesterol synthesis pathway in all PDGCs, its inhibition is only particularly sensitive in the OXPPOS subtype PDGCs (Supplementary Fig. 6a–d). However, subtyping methods based on the patient transcriptome cannot predict cellular sensitivity to lovastatin (Supplementary Data 8). The current scope of our drug sensitivity experiments is mainly focused on investigating the relationship between tumor plasticity and therapeutic response. To fully investigate the relationship between tumor subtypes and drug resistance, in-depth exome sequencing in conjunction with longer treatment duration and refined treatment assays would be requisite.

We found that the cholesterol-lowering drug lovastatin prevents the growth of OXPPOS subtype PDGCs in vitro by inhibiting cellular respiration. Oral administration of lovastatin inhibited the growth of subcutaneous tumors but was ineffective in intracranial xenografts of GBM. This finding is consistent with numerous animal and clinical studies showing that various statins are virtually incapable of altering intracranial or cerebrospinal fluid concentrations of cholesterol, regardless of the duration of administration and regardless of their ability to cross the blood-brain barrier⁴⁸. The mechanism responsible for this phenomenon remains unclear. It is probably due to the fact that cholesterol is essential for brain development and normal physiological function, as the organism has developed several mechanisms to ensure the stability of cholesterol concentrations^{49–51}. Thus, although the brain is a cholesterol-rich organ, it rarely exchanges cholesterol with the periphery. At the same time, the neurons and glia

in the brain use the Kandutsch–Russell and Bloch pathways for cholesterol synthesis, respectively, to ensure a robust system⁵¹.

We used WGS to identify genomic alterations of PDGCs, which allowed us to analyze ecDNA-related alterations that had previously been overlooked. In recent years, increasing attention has been paid to the role of ecDNA in GBM, and GBM cells have been used as models for ecDNA research^{52–54}. For example, we found that the widely used MGG8 cell line contains more than 50 copies of *MYC*-ecDNA. Previous identification of PDGCs has mainly focused on low-passage stages based on the assumption that the PDGC culture system can stably maintain the genome characteristics⁸. Compared with conventional GBM cell lines cultured in serum-containing medium, PDGCs cultured in a serum-free medium better maintained the intratumoral heterogeneity and transcriptional patterns of the parental GBM tissue. However, this culture system is not without flaws. We observed a high proportion of PDGCs with *MYC/MYC*N amplification, which is rare in GBM tissues. The occurrence of *MYC/MYC*N amplification is closely related to the PDGC culture system and does not occur in serum-grown GBM cells. We confirmed that this serum-free culture system can activate the *MYC* pathway and found that a considerable number of GBM tissues contain rare tumor cells with *MYC* amplification, which have some growth advantage in the serum-free culture system and may eventually take over the entire culture system.

In our study, most MES subtype PDGCs failed to form intracranial tumors in immunodeficient mice. Xie et al. also reported that only 7 of 17 MES subtype GSCs could form intracranial tumors¹⁹. Therefore, most in vivo studies focusing on the development and treatment of GBM may be mainly limited to the PN and OXPHOS subtypes, while the MES subtype is relatively understudied. In addition, our drug screening failed to identify any compounds specific to the MES subtype. The presence of multiple tumor cell subtypes in a single GBM tissue and the inability to effectively prevent tumor development by eliminating only the tumor cells of PN and OXPHOS subtypes may be one of the reasons for the low success rate of translational research from basic GBM research to clinical practice.

Methods

Collection and Processing of Patient-derived Glioma Cells (PDGCs)

All the patients involved provided written informed consent. All the sampling steps and experimental procedures in this study were approved by the ethics committee of Beijing Tiantan Hospital, Capital Medical University (KY 2020-093-02 and KY 2014-021-02). Sex and gender were not considered in this study. No compensation was provided to the participants.

In total, we collected 12 PDGCs from other laboratories and established 38 patients-derived (37 primary GBM and one recurrent GBM) glioma cell lines cultured in serum-free medium. Specifically, we successfully cultured 28 PDGCs from a total of 86 glioma specimens, resulting in an overall success rate of 32.6%. Of these, 12 PDGCs were successfully established from 32 GBM samples collected from hospitals in Suzhou and Nanjing. The remaining 16 PDGCs with the prefix ‘CB’ were established from 54 GBM samples from Tiantan Hospital in Beijing. Of these 16 cell lines, 12 were subsequently used for comparative analysis with the RNA sequencing data from their original tumor tissues.

Briefly, fresh glioma tissues were immediately placed in Hypo ThermoSol FRS (Biolife Solutions, 101104) after surgery resection. The tissues need to be processed as quickly as possible because the longer the time between surgical removal and tissue processing, the more difficult it is for the PDGCs to survive. The tissues were transferred to a sterile tube and minced with dissection scissors in a biosafety cabinet. Accutase cell detachment solution (BioLegend, 423201) was added to the tube to resuspend the tissue pieces. To adequately disassociate tissues into single-cell suspension, the tube with the tissue pieces was

incubated in a 37 °C-water bath for 5–10 min. Subsequently, the suspension in the tube was washed with cold PBS through a 40 µm cell strainer (Falcon, 352340) to remove un-disassociated tissue pieces. The cell precipitation was obtained by centrifuge at 500 × *g* for 5 min at 4 °C. The supernatant was removed, and red blood cell lysis buffer (Beyotime, C3702) was added with gentle pipetting at room temperature for 3–5 min. Then centrifuged to remove the red supernatant. Washed the cells twice with cold PBS. The cells were resuspended in a serum-free culture medium and seeded into cell culture dishes.

Cell culture

For the first two passages of newly established PDGCs, cells were cultured as a spheroid. After that the PDGCs were seeded into cell culture dishes (NEST, 705001), which were coated by Matrigel (CORNING, 354248), and cultured in PDGC serum-free medium containing DMEM/F12 (Gibco, C11330500BT), 1% penicillin/streptomycin (Solarbio, P8420/S8290), 1× B-27 without vitamin A supplement (Gibco, 12587010), 1× N-2 supplement (Gibco, 17502048), 0.5% glutaMAX-1 (Gibco, 35050-061), 5 mM HEPES (Beyotime, ST090), 600 µg/mL Glucose (Sangon Biotech, A501991-0500), 50 µM 2-mercaptoethanol (SIGMA, M3148-100ML), 20 ng/mL EGF (Novoprotein, C029-500 µg), and 20 ng/mL bFGF (ORIGENE, TP750002) and placed within a 37 °C, 5% CO₂, 5% O₂, and 90% humidity sterile incubator. Usually, half of the medium was replaced every two days. When fully confluent, the PDGCs were split 1:2–1:3 ratios by Accutase cell detachment solution. In the early stage (<10 passages), PDGCs were frozen in each passage to preserve the original cell samples and expanded for further studies. All PDGCs in this study could be passed on for 10 generations or more. The term low-passage or long-passage in the study refers to the number of passages of cells when the cells were harvested for analysis.

DNA and RNA extraction

For tissue samples from glioma patients, fresh-frozen tissue RNA was extracted by FastPure Cell/Tissue Total RNA Isolation Kit (Vazyme, RC112-01) according to the manufacturer’s instructions. For the PDGC sample, 5 × 10⁶–1 × 10⁷ cells were collected per cell line for DNA and RNA extraction. DNA of the PDGCs was extracted by TIANamp Genomic DNA Kit (TIANGEN, DP304), and RNA was extracted using standard TRIzol RNA extraction (Invitrogen, 15596-026). All nuclear samples were quantified using the NanoDrop instrument.

Small molecule drugs library screening assay

The small molecule library (Selleck) included 1466 small molecule drugs approved by the FDA with a stock concentration of 10 mM in DMSO or water. These compounds were added to individual 96-wells at a concentration of 10 µM to select which of them would inhibit cell viability. This round of screening used three PDGCs (G98, G709, and G118). Small molecules with cell viability inhibition (cell viability below 0.25) in at least one cell line were selected for the second round of screening. In the second screening round, 41 PDGCs were tested at a concentration of 5 µM. Cell viability was determined using CellTiter-Glo (Promega, G9243).

Cell viability assay

1500 cells per well were seeded in 96-well plates. After 4–6 h of incubation in a 37 °C incubator, cells were treated with different small molecule compounds as described in figure legends for three days. Cell viability was measured with CellTiter-Glo (Promega, G9243) after the treatment. Small molecule compounds purchased from Selleck included lovastatin (S2061) and cholesterol (S4154), and from Shanghai yuanye Bio-Technology included lanosterol (S27466).

Immunofluorescence (IF)

For mitochondria staining, BNI423 cultured on glass coverslips was incubated with lovastatin with/without cholesterol for 16 h.

Subsequently, cells were incubated with 200 nM MitoTracker Deep Red FM (Invitrogen, M22426) for 40 min at 37 °C, then washed in pre-heated PBS. Samples were fixed in 4% paraformaldehyde (PFA) at room temperature for 1–5 min and then permeabilized with PBST (PBS with 0.2% Triton X-100) for 10 min. Finally, samples were mounted with an antifade mounting medium with DAPI (Beyotime, P0131). Zeiss LSM880 confocal microscope with Airyscan was used to acquire images on a 63× oil objective.

ADP/ATP ratio assay

For the cellular ADP/ATP ratio of BNI423, 3,000 cells/well were seeded in 96-well plates. After overnight incubation in a 37 °C incubator, the small molecule compounds were added to cultured cells. ADP/ATP Ratio was performed by ADP/ATP Ratio Assay Kit (SIGMA, MAK135) according to the manufacturer's instructions after treatment.

Apoptosis assay

For Annexin V staining, BNI423 cells were seeded in 6-well plates with a density of 20%–30%. After overnight incubation, cells were treated with lovastatin or vehicle (DMSO) for 12, 24, 48, and 72 h. FITC Annexin V/Propidium Iodide (PI) staining was performed by FITC Annexin V Apoptosis Detection Kit with PI (BioLegend, 640914). Flow cytometry was performed on the BD LSR Fortessa (BD Biosciences) instrument, and data were analyzed using FlowJo (version 10.7.1).

In vivo test of lovastatin

Female BALB/c Nude mice (6–8 weeks) were purchased from Vital River and housed in the laboratory animal resource center at the Chinese Institute for Brain Research (CIBR), Beijing. The mice were housed at 21.5 °C, 50% humidity, and a 12-h dark-light cycle. All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee at CIBR. The maximum tumor diameter allowed by the CIBR Ethics Committee was 20 mm, and the weight of the tumor must not exceed 10% of the body weight of the mouse. None of the tumor volumes of mice in this study exceeded the regulations.

For the subcutaneous PDGCs' xenograft model, 1,000,000 BNI17 (in matrigel) were subcutaneously injected into the left flanks of nude mice. For drug treatment studies, lovastatin (50 mg/kg) or vehicle (PBS) was administered to mice by intragastrical (i.g.) daily after the tumor volume reached about 50 mm³. Tumor volume (V) was calculated with the formula $V = (\text{length (L)} \times \text{width (W)}^2) \times 0.5$, where length and width were measured with the vernier caliper daily. Animals were observed until the tumor volume exceeded the limits (1000 mm³). For the intracranial orthotopic PDGCs' xenograft model, 500,000 BNI423 (in 5 µL matrigel) was transplanted stereotactically into the left striatum of nude mice by Standard Stereotaxic Instruments (RWD). The coordinates of PDGCs injection were +0.5 mm AP, −1.9 mm ML, and −3.6 mm DV with respect to the bregma, and PDGCs were injected with a microsyringe (Hamilton, 701). Five weeks after the PDGCs transplantation, lovastatin (3.5 mg/mL) or vehicle was injected into the tumor site through the implanted brain infusion cannula. The brain infusion cannula (RWD, BIC-5) was connected via a 1.5 cm-long tubing to a subcutaneous osmotic pump (RWD, 1001 W) to deliver the lovastatin or vehicle (flow rate 0.5 µL/h). After a three-day treatment, mice were perfused and whole brains were excised for further analysis.

Histology and immunohistochemistry (IHC)

The tissue was fixed in cold 4% PFA (Sangon Biotech, A500684-0500) overnight. The following day, fixed tissues were dehydrated through graded ethanol and embedded in paraffin. Paraffin blocks were sectioned at 3 µm intervals using a paraffin microtome (SLEE, Mainz, Germany) for H&E/IHC. The slides were deparaffinized and used the standard H&E staining protocol for histology. For IHC, antigen retrieval was performed in sodium citrate buffer (pH 6.0) in the pressure cooker. After washing three times with ddH₂O, the slides were incubated in

3% H₂O₂ (Aladdin, H112515) for 10 min at room temperature. Then washing with ddH₂O again and followed by blocking 3% BSA (SIGMA, V900933-100G) for 1 h at room temperature. Primary antibodies used included Ki67 (ORIGENE, TA802544) and cleaved caspase-3 (Cell Signaling Technology, 9661S). After overnight incubation of primary antibodies at 4 °C, the slides were washed 4–5 times with Tris Buffered Saline with Tween 20 (TBST). Chromogenic detection was performed with a two-step HRP-Polymer detection kit (ZSGB-BIO, PV-8000-1) and DAB kit (ZSGB-BIO, ZLI-9018). Olympus VS120 and ZEISS Axio Observer7 were used to acquire images and HALO was used for histopathological analysis.

Fluorescence in situ hybridization (FISH)

FISH was performed on formalin-fixed paraffin embedding original GBM tissues. After the selection of an area containing more than 70% tumor cells, 3 µm sections were deparaffinized, and dehydrated with ethanol. Hybridization was performed according to the manufacturer's instructions. *MYC* (8q24) gene amplification probe (LBP, F.01006) was purchased from Guangzhou LBP Medicine Science and Technology Co., Ltd, with chromosome 8 labeled by the centromere-specific probe. *MYCN* (2p24) gene probe (LBP, F.01013) was used to label *MYCN*, with a control probe to label *LAF4*. We determined a sample tissue carrying *MYC/MYCN* amplification if there were more than three cells with *MYC/MYCN* amplification in the tissue.

Quantitative PCR (qPCR)

For quantitative PCR, cDNA was generated using HiScript III All-in-one RT SuperMix Perfect for qPCR (Vazyme, R333). Quantitative PCR analysis was performed with Taq Pro Universal SYBR qPCR Master Mix (Vazyme, Q712) using a CFX96 Real-time System (Bio-Rad). Transcript quantities were determined using the difference of the C_q method, and values were normalized to the expression of *GAPDH*. Primer sequences were as follows:

GAPDH-F: 5'-AATCCCATCACCATCTTCCA-3',
GAPDH-R: 5'-TGGACTCCACGACGTACTCA-3',
HMGS1-F: 5'-GATGTGGGAATTGTTGCCCTT-3',
HMGS1-R: 5'-ATTGTCTCTGTTCACACTCCAG-3',
HMGR-F: 5'-TGATTGACCTTTCCAGAGCAAG-3',
HMGR-R: 5'-CTAAATGCCATTCCACGAGC-3',
SQLE-F: 5'-GGCATTGCCACTTTCACCTAT-3',
SQLE-R: 5'-GGCCTGAGAGAATATCCGAGAAG-3'.

Droplet digital PCR

Droplet digital PCR (ddPCR) was performed on 20 PDGCs' genomic DNA to estimate the copy number of *MYC* and *EGFR*. To contain ecDNA, genomic DNA was isolated from 20 PDGCs using ethanol precipitation. The reference gene, *HMBS*, and the calculation of copy number were referred to in previous literature⁵⁵. The *EGFR* and *HMBS* primers were also referenced in the literature⁵⁵ and the *MYC* primers were self-designed, the sequences were as follows:

HMBS-F: 5'-GGGACAGTGATCCCAAGGTC-3',
HMBS-R: 5'-CTGAGGTAACCGATCTGACG-3',
EGFR-F: 5'-CGAAGGAAAGTAAGGAGCTG-3',
EGFR-R: 5'-AAGAAGATGCACGAAGGC-3',
MYC-F: 5'-GGACTATCCTGCTGCCAAGA-3',
MYC-R: 5'-CGCCTCTTGACATTCTCTC-3'.

Reactions (20 µL) were prepared with 24 ng of gDNA, QX200™ ddPCR™ EvaGreen Supermix (Bio-Rad, 1864034), and 100 nM primers. Micro droplet generation, PCR reaction, and machine operations were performed according to the manufacturer's protocol.

Bulk RNA-seq data analysis

Sequencing adapters of raw reads were trimmed by fastp⁵⁶ (version 0.23.1), and the resulting clean reads were mapped to the hg38 reference genome with STAR⁵⁷ (version 2.7.3a). Quantification of gene

expression level was conducted by featureCounts⁵⁸ (version 2.0.0). The final gene expression matrix was normalized by counts per million (CPM).

Bulk WGS/WES data analysis

Clean reads were aligned to the hg38 reference genome using bwa mem algorithm⁵⁹ in Sentieon (Sentieon Inc, San Jose, CA). Germline mutations were called following the DNaseq pipeline in Sentieon (https://support.sentieon.com/manual/DNaseq_usage/dnaseq/), and somatic mutations were called following the TNseq pipeline in Sentieon (https://support.sentieon.com/manual/TNseq_usage/tnseq/). Mutations were annotated by annovar⁶⁰ (version 2019Oct24). Copy number variations (CNV) called by both CNVnator⁶¹ (version 0.4.1) and hmftools-PURPLE (version 2.51) were retained for further analysis. In addition, extrachromosomal DNA (ecDNA) was analyzed by AmpliconArchitect⁶² (version 2019).

Subtype determination based on non-negative matrix factorization

We utilized the non-negative matrix factorization (NMF) clustering method to identify different subtypes among 50 glioma cell lines. Briefly, genes were ranked by the expression values' median absolute deviation (MAD). We sorted MAD values from high to low and selected the top-rank genes for NMF clustering. We tested a range of ranks between 2 and 7. When 50 glioma cell lines were clustered into four subtypes with the top 4500 genes, we received the highest cophenetic scores. To generate stable clustering results, we used the "leave one out" method. Concretely, we only used 49 samples for NMF clustering each time. For example, we left the first sample out and used the 2nd to 50th samples for clustering, and the next time, we left the second sample out and used the remaining samples for clustering. Finally, we obtained 50 clustering results (Supplementary Fig. 2b). Given the fact that samples within the same subtype tended to cluster together, we determined the clustering identity based on 50 results.

Identification of subtype-specific signatures

Subtype-specified signatures were identified²². For gene i , we divided the expression values into two groups: A (what we are interested in) and B (the remaining samples). We defined a normalized enrichment score, NES, to estimate the level that the expression of gene i in group A is greater than the expression of gene i in group B.

$$NES = \frac{T - \frac{m(m+1)}{2}}{mn} \quad (1)$$

Where m is the number of samples in group A, n is the number of samples in group B, and T is the sum of ranks of gene i expression in group A. Next, we transformed NES into a fold change-based score. The higher the score is, the higher the expression of gene i in group A.

$$score = \log 2 \left(\frac{NES}{1 - NES} \right) \quad (2)$$

To identify the subtype-specific signatures, we calculated FC as follows:

$$FC = \log 2(E_A/E_B) \quad (3)$$

where E_A is the average expression of gene i in group A, and E_B is the average expression of gene i in group B. P -values were calculated by two-sided Wilcoxon rank-sum test, and adjusted p -values by FDR. The signature of group A was determined by the following criteria: (1) $FC > 1.5$; (2) adjusted $p < 0.01$; (3) intersection with GBM-intrinsic genes; (4) top 100 ranked (from high to low) score.

scRNA-seq data analysis

Couturier et al. data²⁵ was downloaded from https://github.com/mbourgey/scRNA_GBM. Basic analysis was performed by the R package Seurat⁶³ (version 4.0.3). Cells meeting the following criteria were retained for downstream analysis: (1) the number of detected genes was between 800 and 7500; (2) the number of unique molecular identifiers was between 1500 and 50,000; (3) the percentage of mitochondrial counts was less than 10%; (4) the percentage of ribosomal counts was less than 30%. After quality control, cells from different patients were integrated by riger⁶⁴ (version 1.0.0). Cells were annotated by canonical markers as indicated by the original article. *MCAM* and *ESAM* were expressed by endothelial cells, *CD53* and *CD68* were expressed by myeloid cells, *MOG* and *MBP* were expressed by oligodendrocytes. Transcriptome-based CNV inference was performed by R package infercnv⁶⁵ (version 1.6.0). Endothelial cells, myeloid cells, and oligodendrocytes were considered as references, with parameters "cutoff=0.1, denoise=TRUE, HMM=TRUE". The resulting file run.final.infercnv_obj input to ComplexHeatmap⁶⁶ (version 2.6.2) for the visualization of CNV levels. Neftel et al. data¹⁸ were downloaded from https://singlecell.broadinstitute.org/single_cell. Data were processed as above.

DNA methylation data analysis

DNA methylation data were downloaded from GEO with accession number GSE92462, where the DNA methylation level was measured by Infinium 450k. The IDAT files were read by R package minfi⁶⁷ with the command read.metharray.exp, then preprocessed by preprocessIllumina. The copy number was estimated by R package conumee⁶⁸, with neural stem cells as control.

Transcriptional similarity between PDGCs and their parental tissues

To determine the similarity between PDGCs and their derived GBM tissues, we used the RNA-seq data of 12 matched PDGC-tissue and obtained 18,278 common genes between PDGCs and GBM tissues. We then overlapped these genes with GBM-intrinsic genes defined by Wang et al.¹⁷, and obtained 6780 genes. We constructed an expression matrix, with rows for selected GBM-intrinsic genes, and columns for 12 PDGCs and 12 parental tissues. Next, we calculated the expressional correlation of these genes between PDGCs and GBM tissues, and the top 1000 most correlated genes were selected for transcriptional similarity analysis. The correlation of samples was calculated and visualized by R package ggcorrplot (version 0.1.4.1).

To eliminate the influence of non-tumor cells within the bulk tumor RNA-seq data, we used ESTIMATE (version 1.0.13) to calculate tumor purity and retained samples with tumor purity greater than 90%, obtaining five pairs (CB5304, CB5399, CB2065, CB7936, and CB1838) of PDGC and matched tissue samples. We selected the top 1000 variable genes evaluated by median absolute deviation (MAD) and subsequently applied PCA to the expression profile of the five PDGC-tissue paired samples. We also quantified the distances of the five paired PDGC-tissue samples and the unpaired samples. For example, the unpaired samples of PDGC #1 include PDGC #2, #3, #4, #5 and tissue #2, #3, #4, #5, and so forth, resulting in a total of 40 possible unpaired samples.

Genome alterations similarity between PDGCs and TCGA GBM patients

We used WGS data of PDGCs to determine nonsynonymous mutations and copy number variations (CNVs) and summarized the gene alteration frequency. We also downloaded the mutations and CNVs data of GBM patients from TCGA (<http://xena.ucsc.edu/welcome-to-ucsc-xena/>), of which 393 patients with both mutations and CNVs information were retained for downstream analysis. We selected the top 30

most frequently altered genes in TCGA GBM patients and correlated their alteration frequencies of these genes with those in PDGCs.

Calculation of subtype score

All the subtype signatures involved in this study were summarized in Supplementary Data 2. Subtype score was calculated by R package ssGSEA^{69,70}. Briefly, the ssGSEA score of each sample (for bulk RNA-seq) or cell (for scRNA-seq) was calculated and normalized to values between 0 and 1. In addition, we performed 1000 permutations for each subtype signature and got the corresponding *p*-value. The subtype identity of each sample or cell was determined by the smallest *p*-value.

Robustness Evaluation of Subtyping System

To evaluate the performance of our classification system in determining subtypes across PDGCs, we performed an in silico mimicry experiment based on the expression profiles of PDGCs (50 PDGCs in total). First, PDGCs with indeterminate subtypes were excluded (see Supplementary Data 1), leaving 45 PDGCs for subsequent analysis. For each target PDGC, we randomly selected a background PDGC from the pool of remaining PDGCs and generated a mock expression profile of PDGC by blending the target PDGC with different proportions (e.g., 10%, 20%, 30%, and 40%) of the background PDGC, resulting in a total of 180 mock PDGC expression profiles. We then determined the subtype identity based on these mimicries. Robustness was calculated based on the percentage of subtype consistency between the target PDGC and the mock PDGC.

Drug response data analysis

We initially used a drug library composed of 1466 FDA-approved small molecules to treat three PDGCs (G98, G118, and G709) respectively. After three days of exposure, we compared the cell viability between PDGCs with or without drug treatment. If the drug treatment (with a dose of 10 μ M) inhibited cell viability (i.e., $\frac{\text{cell viability in PDGCs with drug treat}}{\text{cell viability in PDGCs with vehicle treat}} < 25\%$), this drug was kept for further analysis. We obtained 214 small molecules that had the potential ability to inhibit glioma cell growth. Subsequently, we used these 214 drugs (with a dose of 5 μ M) to treat 41 PDGCs for three days respectively, and measured the cell viability upon drug treatment. We defined normalized cell viability (i.e., $\log_2 \frac{\text{cell viability of PDGCs with drug treat}}{\text{cell viability of PDGCs with vehicle treat}}$) to measure the inhibitory effect of a drug on PDGCs. The lower the normalized cell viability, the stronger the inhibitory effects of the drug on the cells. Heatmap of the drug response data was generated by R package ComplexHeatmap⁶⁶ (version 2.6.2).

Gene set enrichment analysis

To detect the subtype-specific pathways, we download pathway information from MSigDB v7.3 (<https://www.gsea-msigdb.org/gsea/msigdb/>). Gene set enrichment analysis was performed by R package fgsea (version 1.16.0), with genes ranked by expression fold change. Visualization of enriched pathways was conducted by ggplot2⁷¹ (version 3.3.5).

Survival analysis

The survival data of TCGA patients were downloaded from UCSC Xena (<http://xena.ucsc.edu/welcome-to-ucsc-xena/>). The subtype of patients in the TCGA cohort was determined as above. Kaplan–Meier survival curves were generated by the R package survminer (version 0.4.9) to estimate the overall survival difference between subtypes, and *p*-values were calculated using the log-rank test.

Statistics & reproducibility

Detailed statistical information was listed in respective figure legends or methods. Asterisks are used to indicate the statistical significance

(**p* < 0.05, ***p* < 0.01, ****p* < 0.001); n.s. means statistically non-significant (*p* > 0.05).

No statistical method was used to predetermine the sample size. No data were excluded from the analyses. Group allocation was performed in a randomized fashion. The investigators were not blinded to allocation during outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Couturier et al. data²⁵ was downloaded from GitHub (https://github.com/mbourgey/scRNA_GBM); Neftel et al. data¹⁸ was downloaded from Single Cell PORTAL (https://singlecell.broadinstitute.org/single_cell); DNA methylation data of PDGCs were downloaded from Gene Expression Omnibus (GEO) database by accession number GSE92462; Segerman et al.²⁶ data and Mack et al.²³ data were obtained from GEO database by accession numbers GSE89399 and GSE119834, respectively; HGCC data were obtained from <http://portal.hgcc.se/#>; WGS data of CCLC cell lines were downloaded from Sequence Read Archive (SRA) database, accession number PRJNA523380; expression profiles, mutations, CNVs, and survival data of TCGA GBM patients were downloaded from UCSC Xena (<http://xena.ucsc.edu/welcome-to-ucsc-xena/>). All in-house established glioma cell lines in this study are available to research groups with a standard materials transfer agreement and can be obtained by contacting the corresponding author. The raw RNA-seq, WES, and WGS data generated in this study have been deposited in the Genome Sequence Archive (GSA) at the National Genomics Data Center, China National Center for Bioinformatics/Beijing Institute of Genomics, Chinese Academy of Sciences under accession number HRA003009. In accordance with the regulations on the management of human genetic resources by the Minister of Science and Technology, HRA003009 is accessible in a controlled manner. Data are held and distributed by the GSA for Human. The guidelines for data requests can be found at <https://ngdc.cncb.ac.cn/gsa-human/document>. There will be no restrictions for non-profit research organizations. The Data Access Committee will provide feedback on the request within one week and inform the GSA for Human whether access to the data will be granted. Access can be granted by fulfilling the institute's data transfer agreement. The processed count matrices, genomic alterations, and drug response data have been deposited on our laboratory website (<https://chenlab.cibr.ac.cn/kxyj/sjgk/index.htm>). All other data are available in the article and its Supplementary files or from the corresponding author upon request. Source data are provided with this paper.

Code availability

The source code related to the figures of the study is available on GitHub (<https://github.com/CIBRChenLab/Patient-derived-Glioma-Cell-Lines>, <https://doi.org/10.5281/zenodo.12532081>).

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Author contributions

M.W., R.Y., and J.C. conceptualized and designed this study. M.W. and J.C. wrote the manuscript. P.C., J.R.Z., and Y.L. constructed the PDGC lines. J.L. tested the tumorigenic ability of PDGCs. T.W. performed the initial drug screening and experiments during the paper revision. L.W. contributed to the paper revision. M.L. performed the drug screening experiments. R.Y. validated the effects of lovastatin in vitro and in vivo. M.W. performed bioinformatic analysis, drug response analysis, and generated figures. N.J., T.L., J.X.Z., G.L., Y.T., Z.G., X.W., A.W., W.C., M.G., G.C., R.L., Y.Y., and W.Z. provided the human glioma tissues or cell lines. Y.Y., W.Z., Q.W., and J.C. supervised the project and reviewed the manuscript. J.C. acquired financial support.

Competing interests

The authors declare no competing interests.

Additional information

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